

VULNERABILITY ASSESSMENT REPORT FOR A LIVE WEBSITE

**Prepared By:
R. KAMALSRI**

**Internship Domain:
Cyber Security**

**Organization:
Future Interns**

Task:
Task 1 – Vulnerability Assessment Report

**Repository Name:
FUTURE_CS_01**

INTRODUCTION

Cybersecurity plays an important role in protecting websites, applications, and user data from online threats and attacks. Vulnerability assessment is one of the key processes used to identify security weaknesses before they can be exploited by attackers.

This report presents a basic vulnerability assessment conducted on a demo web application designed for cybersecurity learning purposes. The assessment was performed using basic security analysis techniques and tools to identify common vulnerabilities, information exposure issues, and security misconfigurations.

The purpose of this task is to understand how vulnerability assessment works, learn how to classify security risks, and provide suitable remediation recommendations to improve overall web application security.

OBJECTIVES

The main objective of this vulnerability assessment is to identify common security weaknesses present in a web application and analyze their potential impact on users and systems.

This task helps in understanding how cybersecurity professionals perform basic security testing, classify risks based on severity levels, and provide appropriate remediation recommendations to improve website security.

TOOLS USED

- Nmap – Used for network scanning and identifying open ports and services.
- Browser Developer Tools – Used to inspect website elements, cookies, and security-related information.
- Canva – Used for designing and creating the professional vulnerability assessment report.
- GitHub – Used for documentation and project submission.

TARGET WEBSITE

The target selected for this vulnerability assessment is a demo website created specifically for cybersecurity testing and educational purposes.

Website URL:

<https://testphp.vulnweb.com>

This website is intentionally designed to help students and cybersecurity learners practice vulnerability assessment and web security analysis in a safe and legal environment.

METHODOLOGY

The vulnerability assessment was performed using basic cybersecurity testing methods and browser-based analysis techniques. The process involved examining the target website for common security weaknesses and configuration issues.

The assessment included the following steps:

- Identifying active services and open ports using network scanning techniques.
- Inspecting website behavior, cookies, and browser security information.
- Analyzing possible security misconfigurations and information disclosure issues.
- Classifying identified vulnerabilities based on their risk levels such as Low, Medium, and Informational.
- Providing remediation recommendations to improve the overall security posture of the web application.

Findings

1. Missing Security Headers

Risk Level: Low

The website was found to be missing important HTTP security headers such as X-Frame-Options and Content-Security-Policy. Missing headers can make the application vulnerable to attacks like clickjacking and content injection.

Recommendation:

Implement proper HTTP security headers to improve browser-side security protection.

2. Cookie Security Weakness

Risk Level: Medium

Some cookies used by the website were not configured with Secure or HttpOnly flags.
This may expose session information to attackers.

Recommendation:

Enable Secure and HttpOnly flags for all sensitive cookies.

3. Information Disclosure

Risk Level: Low

The website revealed server and technology-related information that could help attackers understand the application environment.

Recommendation:

Reduce unnecessary server information disclosure and configure custom error pages.

4. Open Ports Detection

Risk Level: Informational

Network scanning identified active services running on the target server. Open services may increase the attack surface if not properly secured.

Recommendation:

Close unnecessary ports and regularly monitor exposed services.

Risk Classification Table

Vulnerability	Risk Level
Missing Security Headers	Low
Cookie Security Weakness	Medium
Information Disclosure	Low
Open Ports Detection	Informational

CONCLUSION

The vulnerability assessment helped identify several basic security weaknesses in the target web application. Although no critical vulnerabilities were discovered, the findings demonstrate the importance of implementing proper security controls and secure configuration practices.

This task improved understanding of vulnerability assessment, risk classification, and cybersecurity reporting.